

● HARD

● ADVANCED MATH

Advanced Math

17

10 questions · ~15 min

Q1

ADVANCED MATH

The area of a rectangular banner is 2,661 square inches. The banner's length x , in inches, is 24 inches longer than its width, in inches. Which equation represents this situation?

- A. $0 = x^2 - 24x - 2,661$
- B. $0 = x^2 - 24x + 2,661$
- C. $0 = x^2 + 24x - 2,661$
- D. $0 = x^2 + 24x + 2,661$

$$f(t) = 55t - 2t^2$$

The function f is defined by the given equation. The function g is defined by $g(t) = f(t) + 3$. Which expression represents the maximum value of $g(t)$?

- A. $3 + \frac{55^2}{2}$
- B. $3 + 2\frac{55^2}{4}$
- C. $3 - 2\frac{55^2}{4}$
- D. $3 - \frac{55^2}{2}$

Which of the following functions has(have) a minimum value at -3?

I. $f(x) = -6(3)^x - 3$

II. $g(x) = -3(6)^x$

A. I only

B. II only

C. I and II

D. Neither I nor II

x	$f(x)$
1	a
2	a^5
3	a^9

For the exponential function f , the table above shows several values of x and their corresponding values of

$$f(x)$$

, where a is a constant greater than 1. If k is a constant and

$$f(k) = a^{29}$$

, what is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$fx = 1.84^{\frac{x}{4}}$$

The function f is defined by the given equation. The equation can be rewritten as $fx = 1 + \frac{p}{100}^x$, where p is a constant. Which of the following is closest to the value of p ?

- A. 16
- B. 21
- C. 46
- D. 96

$$M = 1,800(1.02)^t$$

The equation above models the number of members, M , of a gym t years after the gym opens. Of the following, which equation models the number of members of the gym q quarter years after the gym opens?

- A. $M = 1,800(1.02)^{\frac{q}{4}}$
- B. $M = 1,800(1.02)^{4q}$
- C. $M = 1,800(1.005)^{4q}$
- D. $M = 1,800(1.082)^q$

The function f is defined by $fx = a^x + b$, where a and b are constants and $a > 0$. In the xy -plane, the graph of $y = fx$ has a y -intercept at $(0, -25)$ and passes through the point $(2, 23)$. What is the value of $a + b$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$f(x) = 2722^x$$

The function f is defined by the given equation. If $h(x) = f(x) - 4$, which of the following equations defines function h ?

- A. $h(x) = 172^x$
- B. $h(x) = 682^x$
- C. $h(x) = 27216^x$
- D. $h(x) = 2728^x$

$$f(x) = x^2 - 10x + 13$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

A. -130

B. -13

C. $-\frac{23}{2}$

D. $-\frac{3}{2}$

$$f(x) = -500x^2 + 25,000x$$

The revenue $f(x)$, in dollars, that a company receives from sales of a product is given by the function f above, where x is the unit price, in dollars, of the product. The graph of $y = f(x)$ in the xy -plane intersects the x -axis at 0 and a . What does a represent?

- A. The revenue, in dollars, when the unit price of the product is \$0
- B. The unit price, in dollars, of the product that will result in maximum revenue
- C. The unit price, in dollars, of the product that will result in a revenue of \$0
- D. The maximum revenue, in dollars, that the company can make